Data from MD simulations of simple shear

- simulations from large-scale atomic molecular massively parallel simulator (LAMMPS)[1, 2]

- purely repulsive, binary, 2d system

- $N_A = N_B = 2048$ and $R_A : R_B = 1 : 1.4$

- $NVT = 0$ ensemble

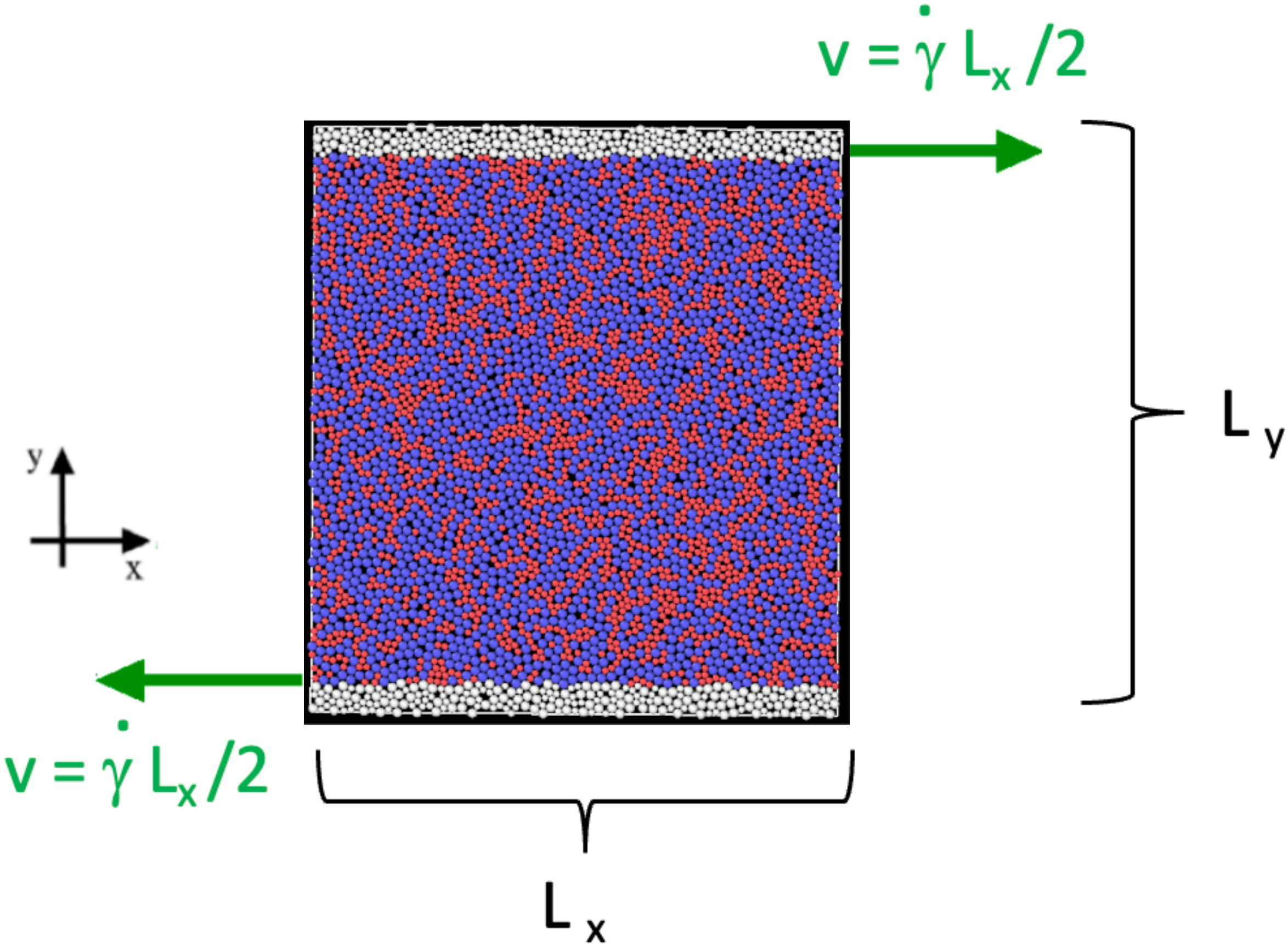
- walls (shown in white):

  - are A and B particles within $6\,R_B$ of boundaries

  - provide quasistatic shear $\dot{\gamma} = 1 \times 10^{-6}$

16

[1]A.P. Thompson et al, Comp. Phys. Comm., 2022.
[2]Plimpton, S. Fast Parallel Algorithms for Short-Range Molecular Dynamics. J. Comput. Phys. 117, 1–19 (1995).


$v = \dot{\gamma} L_x/2$
L_y
y
x
$v = \dot{\gamma} L_x/2$
L_x

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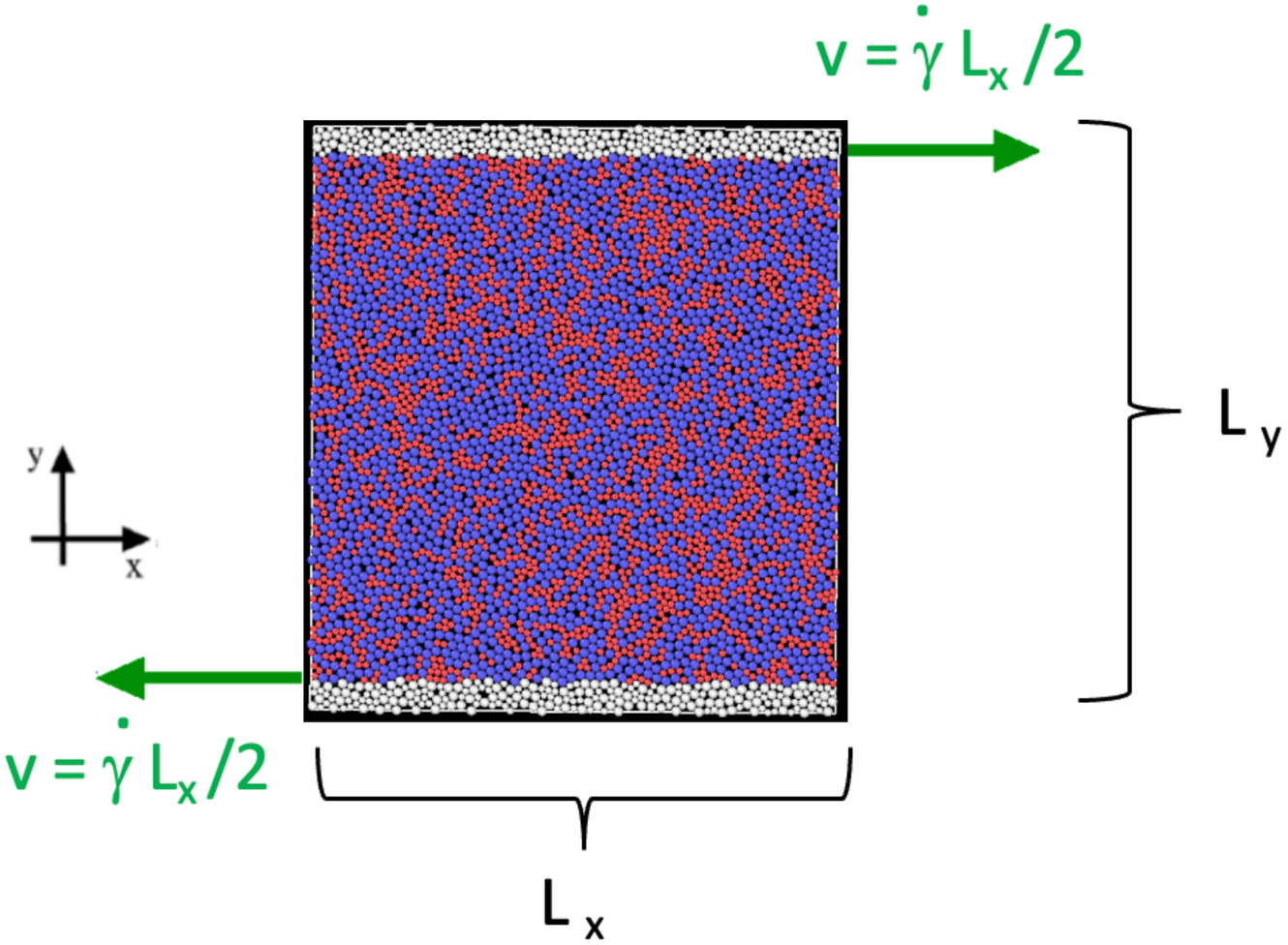

[1]A.P. Thompson et al, Comp. Phys. Comm., 2022.
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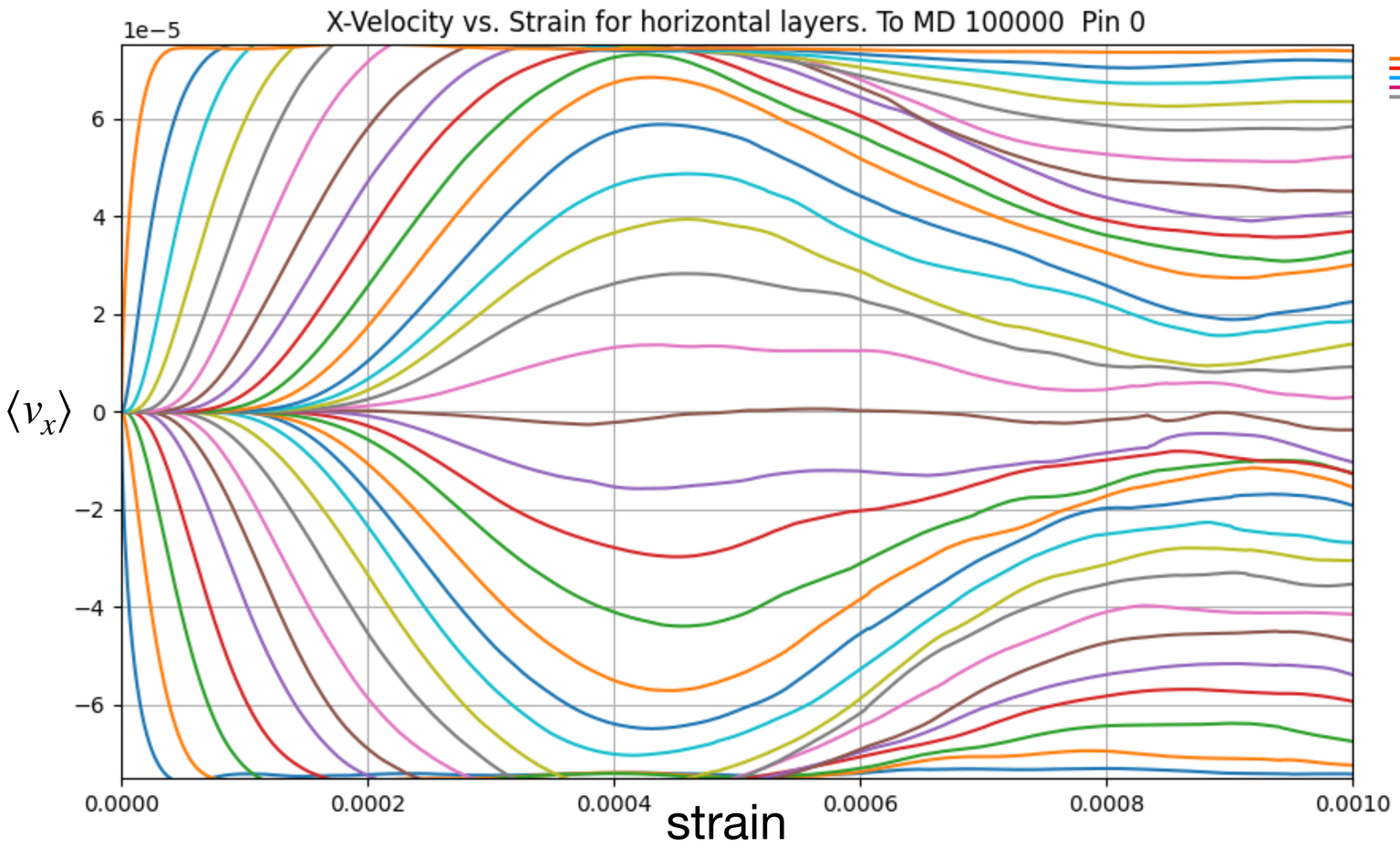
How does the system respond to shearing?

Transient state at early times



Averaged horizontal velocity for 32 layers of particles between walls